Linear Algebra (MT-1004)

Homework#07

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Question 1

Find the Rank of the following matrix

$$A = \begin{bmatrix} 3 & 6 & 9 \\ -1 & 2 & 3 \\ -6 & 12 & -18 \end{bmatrix}$$

Question 2

Given the matrix A:

$$A = \begin{bmatrix} 1 & 3 & 2 & -6 \\ 3 & 9 & 1 & 5 \\ 2 & 6 & -1 & 9 \\ 5 & 15 & 0 & 14 \end{bmatrix}$$

Write a Python program to:

1. Find the basis for the column space (ColA) and the null space (NulA) of the matrix A.
2. Compute and state the dimensions of these subspaces.

Hint: Use NumPy and SymPy libraries for matrix operations and solving systems of linear equations.